Lesson 10.2 - How to Identify Type I and Type II Errors

> **Video:** How To Identify Type I and Type II Errors In Statistics by The Organic Chemistry Tutor **Chapter:** 10 - Hypothesis Testing - 1 Sample

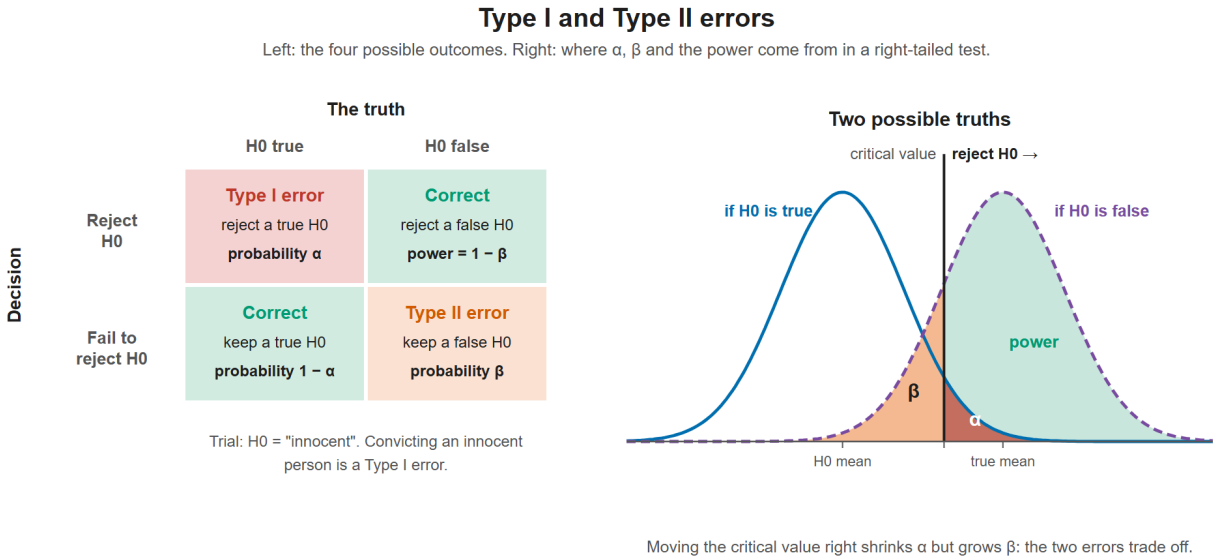
Definitions

Error	What happens	Probability
Type I	**Reject** H_0 when H_0 is actually **true**	α (the significance level)
Type II	**Fail to reject** H_0 when H_0 is actually **false**	β
(correct) Power	Reject H_0 when it is false	$1-\beta$, the **power of the test**

Decision table

	H_0 is **true**	H_0 is **false**
Reject H_0	❌ **Type I error** (α)	✅ Correct (power $= 1-\beta$)
Fail to reject H_0	✅ Correct	❌ **Type II error** (β)

We can't control whether H_0 is true. We only control the decision. (The video sometimes says "accept H_0"; the standard wording is "fail to reject H_0".)



Left: the four combinations of decision and truth. Right: if H0 is true, the red tail beyond the critical value (α) is the chance of a false alarm. If H0 is false, the orange area left of the critical value (β) is the chance of missing the effect, and the green area is the power.

How to identify the error type

1. Mark each scenario as a **good** decision or a **bad** decision (an error).
2. For each error, ask whether the facts make H_0 **true or false**:

- H_0 true but rejected → **Type I**
- H_0 false but not rejected → **Type II**

Example 1 - is John's car safe?

H_0: *John's used car is safe to drive.*

Choice	Situation	Good or bad?	Error
A	Thinks the car is safe, but it is **not** safe	Bad	H_0 false, not rejected → **Type II**
B	Thinks the car is safe, and it **is** safe	Good	—
C	Thinks it may not be safe, and it is **not** safe	Good	—
D	Thinks it may not be safe, but it **is** safe	Bad	H_0 true, rejected → **Type I**

Which error is worse? The Type II error (A): he drives an unsafe car and could be badly hurt. The Type I error (D) only means he stays home unnecessarily.

Example 2 - a criminal trial

H_0: *The defendant is innocent* (presumed innocent).

Choice	Situation	Good or bad?	Error
A	Jury says **guilty**, but he is **innocent**	Bad	H_0 true, rejected → **Type I**
B	Jury says guilty, and he is **not** innocent	Good	—
C	Jury says **not guilty**, but he is **not** innocent	Bad	H_0 false, not rejected → **Type II**
D	Jury says not guilty, and he **is** innocent	Good	—

Which error is worse? Usually the Type I error: an innocent person goes to prison. Letting a guilty person go free (Type II) is also bad, but courts are designed to avoid convicting the innocent. This is partly a judgment call.

Key takeaways

- Type I = **false alarm**: rejecting a true H_0, probability α.
- Type II = **missed detection**: failing to reject a false H_0, probability β.
- Power = $1 - \beta$, the probability of correctly rejecting a false H_0.
- Which error is "worse" depends on the real-world consequences in context.